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Formal Guided Test Sequence Optimization

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Processor Verification - 29 May 2026

Introduction

Verification process faces significant challenges:

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Verification process faces significant challenges:

- # of test sequences grows exponentially with # of test cases.
- Executing all possible sequences is often infeasible
- Identification of the most effective test sequences: crucial for efficient testing.

Agenda

1. Current Verification Approaches
2. Related Works
3. FGTSO Framework
4. Results

Current Verification Approaches

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Formal verification:

Simulation-based verification:

Current Verification Approaches

Formal verification:

- Comprehensive design exploration.
- Encounters state space explosion problem for large scale designs.

Simulation-based verification:

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Formal verification:

- Comprehensive design exploration.
- Encounters state space explosion problem for large scale designs.

Simulation-based verification:

- High scalability and applicability.
- Effectiveness rely on well-designed constraints.
- Biased constraints: miss critical scenarios.
- Strict constraints: increase debugging complexity.

Formal Guided Test Sequence Optimization

- Enhance coverage and verification efficiency.

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- Enhance coverage and verification efficiency.
- Leverage simulation scalability and formal verification precision to target critical conditions
- Integrate simulation and formal verification.
- Introduce abstract models to overcome limitations of BBOX.

Related Works

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RISC-V Torture Test:

RISCV-DV:

Fuzzing tools

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RISC-V Torture Test:

- 👍 Generate random instruction sequences to achieve faster coverage closure
- 👎 Lacks structured verification guidance: limited effectiveness in extreme corner cases

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RISCV-DV:

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Fuzzing tools

- 👎 Cannot guarantee coverage of all possible scenarios.

Problems & Issues

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- False alarms due to environmental discrepancies.
- BBOX issues by excessive complexity or large storage modules.

Formal Guided Test Sequence Optimization Framework

Overall architecture

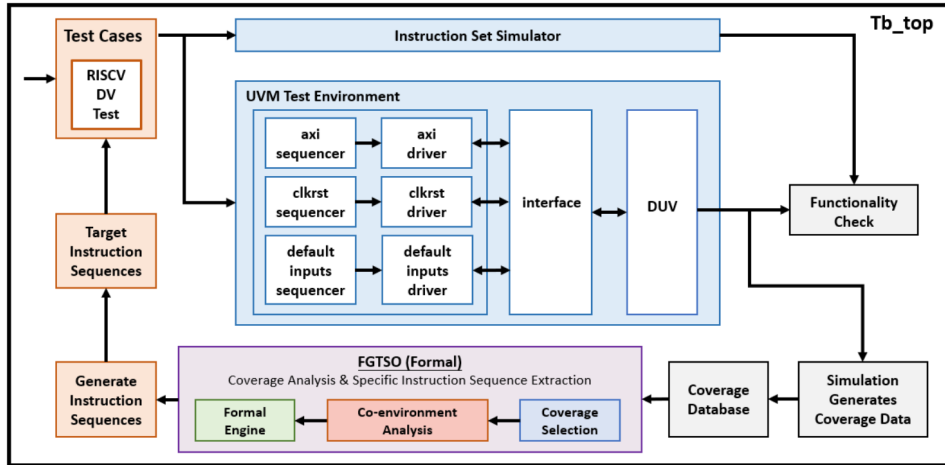


Figure: FGTSO Framework Architecture

Overview

1. Coverage Hole Selection

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2. Environment Consistency Check

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3. Formal Cover Property Generation (*assume* properties)

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4. Targeted Instruction Sequence Generation

FGTSO Workflow

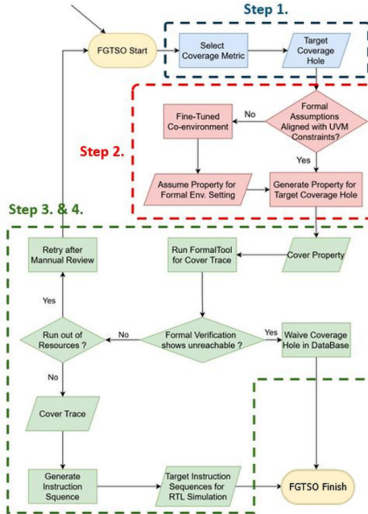


Figure: FGTSO Workflow

FGTSO Workflow - Step 1

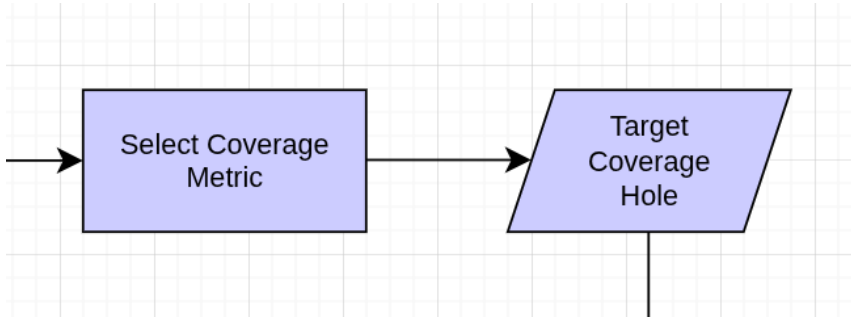


Figure: Step 1: Coverage Hole Selection

FGTSO Workflow - Step 1

Priority: addressing most impactful coverage holes

MaxUncovd strategy

Prioritizes selecting modules with lowest coverage, assuming that addressing the most uncovered areas yields the greatest verification impact.

FGTSO Workflow - Step 2

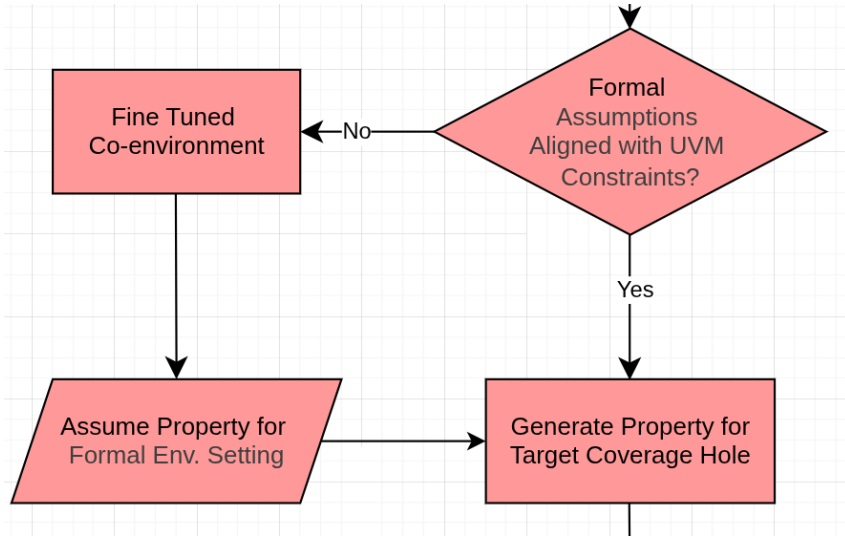


Figure: Step 2: Environment Consistency Check

Coverage Holes Analysis

SCORE	BRANCH	NAME	
86.05	86.05	wt_dcache_ctrl	179 MISS_REQ: begin
87.50	87.50	fpnew_opgroup_fmt_slice	180 miss_req_o = 1'b1;
88.19	88.19	control_mvp	181
88.24	88.24	controller	182 if (req_port_i.kill_req) begin
90.00	90.00	scoreboard	183 req_port_o.data_rvalid = 1'b1;
90.00	90.00	bht	184 if (miss_ack_i) begin
91.67	91.67	fifo_v3	185 state_d = KILL_MISS;
			186 end else begin
			187 state_d = KILL_MISS_ACK;
			188 end
			189 end else if (miss_replay_i) begin
			190 state_d = REPLAY_REQ;
			191 end else if (miss_ack_i) begin
			192 state_d = MISS_WAIT;
			193 end

Figure: Coverage Holes Identification

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- Formal verification generates instruction sequences targeting specific coverage holes, executed in the simulation.
- Results from formal verification and simulation are inconsistent (fail to cover intended coverage holes): misalignment.

FGTSO Workflow Misalignment - PIs

Three are the main causes of misalignment: PIs, BBOX and Insufficient RISC-V-DV constraints.

- Primary Inputs (PIs) (reset, clock, interrupt request, rvalid signal AXI protocol).

FGTSO Workflow Misalignment - Pls

Three are the main causes of misalignment: Pls, BBOX and Insufficient RISC-V-DV constraints.

- Primary Inputs (Pls) (reset, clock, interrupt request, rvalid signal AXI protocol).
 - Assume arbitrary values without constraints.
 - Generated cover traces may fail to address coverage holes.
 - Unconstrained Pls can result in misleading coverage estimation (unreachable states by design)
 - Formal assumptions must be introduced to align formal verification with system behavior.

FGTSO Workflow Misalignment - BBOX Constraints

Problem

Modules with high storage requirements or complex computational logic cannot be fully modeled in formal verification, leading to BBOX constraints.

E.g.: for a cache: huge amount of cache lines. Tracking each line is infeasible.

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Modules with high storage requirements or complex computational logic cannot be fully modeled in formal verification, leading to BBOX constraints.

E.g.: for a cache: huge amount of cache lines. Tracking each line is infeasible.

- Solution: Abstraction models preserve critical behavior while ensuring interaction with the system remains verifiable.
- High Storage Example: Cache abstraction using SVA array to track indices. Predefined tag replaces full tag tracking.
- Computation Example: Symbolic computation approach using SVA.

Cache Abstraction Model

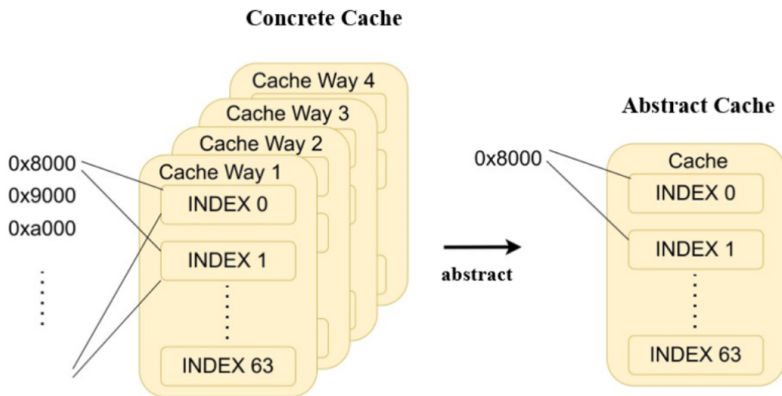


Figure: Cache Abstraction Model

Cache Abstraction Model Codification

```
asm_tag_dcache_idle_0: assume property
(
  if (valid_tag_dcache[gen_cache_wt.i_cache_subsystem.i_wt_dcache.i_wt_dcache_mem.vld_addr])
    ##1 gen_cache_wt.i_cache_subsystem.i_wt_dcache.i_wt_dcache_mem.gen_tag_srams[0].i_tag_sram.rdata_o == 'h100000080000
  else
    ##1 gen_cache_wt.i_cache_subsystem.i_wt_dcache.i_wt_dcache_mem.gen_tag_srams[0].i_tag_sram.rdata_o == 'b0
);
```

Figure: SVA Implementation of Cache Abstraction

FGTSO Workflow Misalignment - RISC-V Constraints

Problem

Formal verification may produce sequences that RISC-V constraints cannot handle.

FGTSO Workflow Misalignment - RISCV-DV Constraints

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Formal verification may produce sequences that RISCV-DV constraints cannot handle.

- Example: Multi-level page table mechanism includes 1 GB, 2 MB, and 4 KB page tables, but original constraints only allow 1 GB page table access.

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Formal verification may produce sequences that RISCV-DV constraints cannot handle.

- Example: Multi-level page table mechanism includes 1 GB, 2 MB, and 4 KB page tables, but original constraints only allow 1 GB page table access.
- Solution: Modify constraints to apply them to addresses pointed to by 1 GB and 2 MB page table entries.

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- Solution: Modify constraints to apply them to addresses pointed to by 1 GB and 2 MB page table entries.
- Implementation: Set X, W, and R bits to 0 to indicate that entries should point to the next level of the page table.

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- Solution: Modify constraints to apply them to addresses pointed to by 1 GB and 2 MB page table entries.
- Implementation: Set X, W, and R bits to 0 to indicate that entries should point to the next level of the page table.
- Result: Coverage holes for multi-level page table accesses can now be filled.

Step 3 & 4: Formal Cover Property Generation

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- Overview: Use formal cover properties to obtain cover traces and generate instruction sequences addressing target coverage holes.
- Property Definition: Step automatically extracts coverage hole expression and applies it to predefined cover templates.
- Formal Execution: Formal tool executes the cover property and derives the input patterns required to satisfy the condition.
- Cover Trace Generation: Generates corresponding cover trace from the formal verification results.
- Sequence Extraction: Extract the instruction sequence from the trace to effectively cover the coverage hole.

Formal Cover Property Example

```
test1: cover property
(
  gen_cache_wt.i_cache_subsystem.i_wt_dcache.gen_rd_ports[1].genblk1.i_wt_dcache_ctrl.state_q == 'h2 &&
  ~gen_cache_wt.i_cache_subsystem.i_wt_dcache.gen_rd_ports[1].genblk1.i_wt_dcache_ctrl.req_port_i.kill_req &&
  gen_cache_wt.i_cache_subsystem.i_wt_dcache.gen_rd_ports[1].genblk1.i_wt_dcache_ctrl.miss_replay_i
);
```

Figure: Cover Property Definition

Handling Coverage Challenges

Handling Coverage Challenges

- Unreachable Coverage Holes: constraints are overly restrictive?
- Waiving Unreachable Holes: If appropriate & hole is unreachable, it is waived; information still logged for debugging.
- Resource Exhaustion: State space explosion.
- Decomposition Strategy: Complex properties decomposed in smaller sub-properties
- Integration: Integrate sub-property traces into comprehensive instruction sequence

Final Outcome

```

main:
    addiw x10, x10, -27
    ld x24, 0(x2)
    slli x8, x8, 8
    addi x8, x2, 0
    csrrw x16, 0x7c1, x2
    lh x15, 4(x2)
    lw x9, 0(x8)
    lhu x3, 1192(x24)
    csrrsi x8, 0x7c1, 3
    lb x28, 1536(x9)
    beq x1, x2, next13632

next13632:
    slli x8, x8, 8
    lbu x0, -2047(x15)
  
```

```

179     MISS_REQ: begin
180         miss_req_o = 1'b1;
181
182         if (req_port_i.kill_req) begin
183             req_port_o.data_rvalid = 1'b1;
184             if (miss_ack_i) begin
185                 state_d = KILL_MISS;
186             end else begin
187                 state_d = KILL_MISS_ACK;
188             end
189         end else if (miss_replay_i) begin
190             state_d = REPLAY_REQ;
191         end else if (miss_ack_i) begin
192             state_d = MISS_WAIT;
193         end
  
```

Figure: FGTSO Results: Coverage Achievement

Comparison with Other Frameworks

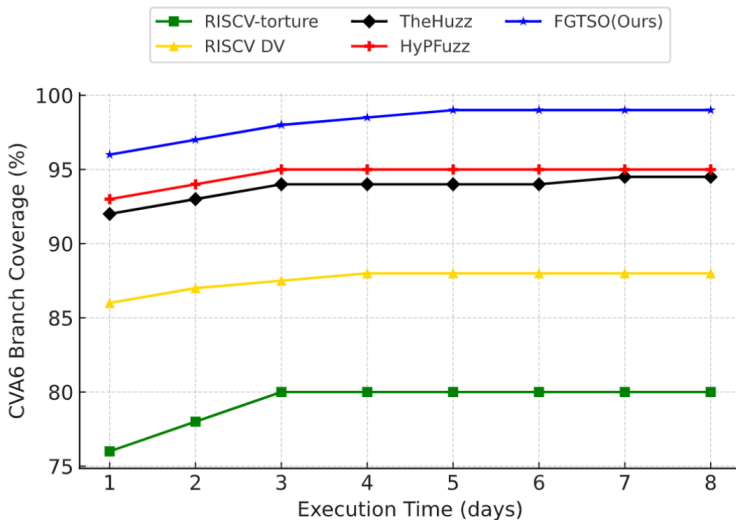


Figure: FGTSO vs. Other Verification Approaches

100% Coverage

Name	Score	Line	Toggle	Condition	Branch
uvmt_cva6_tb	100.00%	100.00%	100.00%	100.00%	100.00%
cva6_dut_wrap	100.00%	100.00%	100.00%	100.00%	100.00%
cva6_tb_wrapper_i	100.00%	100.00%	100.00%	100.00%	100.00%
i_cva6	100.00%	100.00%	100.00%	100.00%	100.00%
commit_sta...	100.00%	100.00%	100.00%	100.00%	100.00%
controller_i	100.00%	100.00%	100.00%	100.00%	100.00%
csr_regfile_i	100.00%	100.00%	100.00%	100.00%	100.00%
ex_stage_i	100.00%	100.00%	100.00%	100.00%	100.00%
gen_cache_...	100.00%	100.00%	100.00%	100.00%	100.00%
i_frontend	100.00%	100.00%	100.00%	100.00%	100.00%
id_stage_i	100.00%	100.00%	100.00%	100.00%	100.00%
issue_stage_i	100.00%	100.00%	100.00%	100.00%	100.00%

Figure: FGTSO: Achieving Complete Coverage

Thanks for your attention!

Questions?

Based on the paper

SHEN, Yu-Shien, et al. Efficient Coverage Optimization with Formal-Guided Testcase Generation in UVM Verification. In: DVCon Europe 2025; Design and Verification Conference and Exhibition. VDE, 2025. p. 43-48.